

Education

University of Utah

B.S. Computer Science

Salt Lake City, UT

August 2020 – May 2025

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies

Associate Software Engineer

Dallas, TX

June 2025 – Present

- Improved throughput and reduced latency for high-frequency defense sensor data pipelines by engineering containerized microservices in C++ and Python on embedded Linux, applying system-level profiling and deterministic scheduling to meet production-grade performance constraints.
- Increased deployment reliability across multiple production targets by building automated CI/CD infrastructure with static analysis and integration testing, enabling consistent validated releases under strict engineering standards.
- Reduced operational overhead for distributed data ingestion by designing fault-tolerant microservice architectures with modular, reusable components that supported rapid scaling across new hardware configurations.

Doxy.me

Software Engineer

Charleston, SC

June 2023 – May 2025

- Reduced clinical review latency by building high-throughput Python and TypeScript backend services processing real-time WebRTC data streams through low-latency inference pipelines deployed on AWS SageMaker and Fargate, sustaining throughput across thousands of concurrent provider sessions.
- Improved system scalability by designing RESTful API contracts and data models that decoupled frontend clients from backend inference infrastructure, enabling independent scaling of each layer under variable load.
- Increased release confidence by owning YAML-based CI/CD configuration and containerized AWS infrastructure automation, achieving zero-downtime deployments across multiple production environments.

University of Utah

Undergraduate Research Assistant – FuTURES Lab

Salt Lake City, UT

May 2024 – Jun 2025

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
 - Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.
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Projects

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

University Course Planning/Review Platform (Class Best): Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.